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Master's Comprehensive Examination

Part I — Theory (90 minutes) — March 26, 1994

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. Suppose n records are manufactured by a particular company. Let X = number among the n that are warped, $p = P$ (warped record).

a. Derive the maximum likelihood estimator of p. If $n = 20$, $X = 3$, what is the estimate?

b. Is the estimator of (a) unbiased?

c. If $n = 20$, $x = 3$, what is the m.l.e. of $(1-p)^4$?

2. Let $X_1, \ldots, X_n$ be observations from a normal distribution with mean μ and variance 1.

a. Find a 95% confidence interval for μ.

b. If n observations are taken, find n large enough so that the length of the confidence interval is .1.

3. Let X be a normal random variable with mean μ and variance 1. Test $H_0 : \mu = 0$ vs $H_1 : \mu > 0$.

a. Indicate the uniformly most powerful test of size .05.

b. Find the power of this test when $\mu = 1$.

4. A card is drawn from a deck of playing cards and then another is drawn at random from those that are left. Find the probability of the events:

a. Neither are hearts.

b. First is heart, second is not heart.

c. Second is a heart.

5. Let X be a random variable whose density is

$$f_X(x) = \begin{cases} cx^2, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise.} \end{cases}$$

a. Find c.

b. Find $P\{\frac{1}{4} < X < \frac{3}{4}\}$.

c. Find the cdf of X.

Master's Comprehensive Examination

Part II Applied (90 minutes) March 26, 1994

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. An experiment was conducted to determine the effectiveness of a new drug in treating hypertension. Ten patients were included in the study. Prior to treatment, the systolic blood pressure was determined for each patient. The patient was administered the new drug for a six-week period. At the completion of the study the systolic blood pressure was measured again. The data from this study are given below:

	Systolic Blood Pressure	
Patient	Baseline	Final
1	160	140
2	180	170
3	150	160
4	150	130
5	170	170
6	180	150
7	200	180
8	140	160
9	190	150
10	180	160

a) Determine if the drug has resulted in a significant *reduction* from baseline in systolic blood pressure. In selecting a statistical method you may assume the data are normally distributed. Be sure to show the null and alternative hypotheses.

b) When *designing* this study the experimenter made one important mistake. Indicate what change you feel is necessary to make this study acceptable to careful review.

2. A calibration curve was developed prior to using an instrument to determine plasma cholesterol. This process consisted of taking five samples with known cholesterol levels (X) and obtaining an instrument reading (Y) for each sample. The result of one such calibration process is given below:

X	Y
100	15.1
150	19.8
200	25.0
250	30.4
300	34.8

Assuming a calibration model of the form

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

answer the following questions:

a) Estimate β_0 and β_1.

b) Test the hypothesis

$$H_0 : \beta_1 = 0, \qquad H_1 : \beta_1 \neq 0.$$

c) Construct a 95% confidence interval for β.

d) Two "unknown" plasma samples are processed on the instrument and give readings of 22.5 and 40.0. For each, estimate the cholesterol level.

e) In (d), which predicted cholesterol level do you believe is better? Why?

3. A three factor factorial experiment was conducted to study the growth of rabbits. The factors included in the experiment were:

Gender – Male, Female

Protein in Diet – Low, Medium, High

Fur Color – Red, White, Black

Four rabbits were placed into each of the 18 treatment combinations. The response variable was the weight gain over a two-month period. A parially completed ANOVA table is given below.

a) Complete the ANOVA table assuming the three factors being studied are fixed effects.

Sources of Variation	Degrees of Freedom	Sum of Squares	Mean Square	Expected Mean Square	F
Total		1500			
Gender		500			
Protein		200			
$G \times P$		50			
Fur		100			
$G \times F$		50			
$P \times F$		30			
$G \times P \times F$		30			
Rabbit ($G \times P \times F$)		540			

b) An analysis of the residuals is also performed. From the analysis it appears that the mean and the standard deviation of the data have a proportional relationship. (Fat bunnies have more variability in their body weight than thin bunnies.) What would you recommend next in the analysis of these data?

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4. To investigate the effect of massive ingestion of ascorbic acid upon food consumption on the proximate body composition of young pigs. Male pigs were force-fed ascorbic acid of 1000mg daily for nine weeks; a control group was included. In Table 1 we present the proximate body compositions (in terms of percent of body weight) of crude lipid in the carcasses of the pigs.

Table 1. Crude Lipid Compositions (%)

Control Group	1000mg Group
23.8	13.8
15.4	9.3
21.7	17.2
18.0	15.1

a) Test the hypothesis that the mean of the control population equals the mean of the 1000mg population. (Do not assume the data are Normally distributed.) Compute the P-value.

b) Determine a 94.2% nonparametric confidence interval for the difference between the two mean populations.

5. In reading Pap smears, pathologists have difficulty in agreeing on whether a certain minor atypical cellular appearance is abnormal or not. Two pathologists classified the same 52 slides independently, grading them zero, indicating normal, to 4, indicating invasive carcinoma. The results are as follows.

		Pathologist A					
		0	1	2	3	4	
	0	21	3	1	0	0	25
Pathologist B	1	1	4	2	1	0	8
	2	0	2	6	4	0	12
	3	0	0	2	2	0	4
	4	0	0	0	0	3	3
		22	9	11	7	3	52

Is there evidence that one pathologist tends to read more pathology than the other? (Test the hypothesis that neither sees more disease than the other. Compute the P-value.)